

few detected at larger angles.

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As the half-life is 33 years, there will be minimal change in the activity of the sample in 2 days, thus activity can be taken as constant for 2 days.

As activity is the number of decay per unit time,

The number of decays in 2 days

= activity x time

= $4.0 \times 10^5 \times 2 \times 24 \times 3600$

= 6.912×10^{10} particles